HR LEAVE MANAGEMENT DASHBOARD

Master Project Documentation & Technical Audit Manual

Power BI | HR Analytics | Employee Leave & Attendance

PROJECT OVERVIEW

This document describes the business purpose, data structure, transformation approach, analytical measures and report interactions for an HR Leave Management Dashboard developed in Power BI. The dashboard is intended to bring employee leave information into a consistent reporting environment and help users review leave activity at employee, department and managerial levels.

The original project outline identifies three principal report areas: Executive Summary, Employee Details and Managerial Hierarchy. This expanded edition retains that structure and adds implementation guidance, validation checks and examples to make the documentation more complete.

Documentation note: *Where the original project outline does not specify an exact field name, business rule or implemented calculation, this document presents a recommended example rather than claiming it is already configured in the Power BI file.*

1. Executive Summary & Project Objectives

Enterprise workforce planning requires visibility into employee availability, leave balances, leave patterns and approval activity. When leave records are maintained across separate files or systems, HR teams may find it difficult to identify trends, compare departments or understand how leave affects team capacity.

The HR Leave Management Dashboard is designed to consolidate employee master information and leave transactions into a single analytical report. Power BI provides the data preparation, relationship modelling and visualization capabilities required to explore the information through filters and interactive visuals.

1.1 Core objectives

- **Centralized attendance tracking:** bring employee leave requests and leave-related information into one reporting view.
- **Operational efficiency:** allow HR users to filter records and inspect individual employee leave activity.
- **Managerial accountability:** support team-level analysis using department and manager reporting structures.
- **Executive visibility:** summarize key indicators such as employee count, leave days and leave distribution.
- **Audit readiness:** make the underlying records and calculation logic easier to review and validate.

1.2 Intended users and value

Stakeholder	Primary information need	Expected value
Executive leadership	Headcount, leave volume and high-level trends	A concise view of workforce leave patterns
HR auditors and operations	Employee-level records, leave categories and data quality	Faster investigation and policy review
Department managers	Team leave patterns and reporting hierarchy	Improved workload and coverage planning

1.3 Scope boundaries

The dashboard focuses on leave and employee analytics. Payroll calculations, statutory compliance decisions, leave encashment payments and automated approval workflows are outside the confirmed scope of the source outline unless separately connected and implemented.

2. Business Requirements & Report Design

The dashboard should present information at different levels of detail. Senior users generally need summary indicators, while HR operations teams need the ability to move from totals to employee records. Managers need a view that reflects their teams and reporting lines.

2.1 Functional requirements

ID	Requirement	Suggested report behaviour
BR-01	View overall leave activity	Show total employees, leave requests and leave days for the selected period.
BR-02	Filter by department	Update relevant charts and tables when a department is selected.
BR-03	Review employee records	Display employee name, department, manager and leave transaction details.
BR-04	Compare leave categories	Break down leave activity by leave type.
BR-05	Review manager teams	Allow users to inspect employees grouped under a manager.
BR-06	Identify data-quality issues	Highlight missing keys, invalid dates and incomplete employee details.

2.2 Suggested report page structure

Report page	Recommended contents	Main audience
Executive Summary	KPI cards, leave trend, department comparison and leave-type distribution	Leadership and HR
Employee Details	Employee table, leave request details and employee-level filters	HR operations and auditors
Managerial Hierarchy	Manager-to-employee matrix, team leave totals and department filters	Department managers

2.3 Usability principles

- Use consistent names, date formats and number formats across all report pages.
- Keep slicers in predictable positions and use clear labels for each filter.
- Use drill-down or drill-through only where it helps users move from summary information to supporting records.
- Make the selected reporting period visible so users understand the time context of each metric.
- Avoid overcrowding a page with too many charts; prioritize the questions each stakeholder needs answered.

3. Data Architecture & Schema Design

The source project outline describes a star-schema-oriented design that separates employee master data from leave transaction records. This separation helps maintain a clear distinction between descriptive employee attributes and individual leave events.

3.1 Core tables

Table	Role	Key fields described in the project
Raw_Employees	Employee master/dimension table	EmployeeID, Employee Name, Department, Job Title, Employment Status, ManagerID
Leave Requests	Leave transaction/fact table	RequestID, EmployeeID, Leave Type, Start Date, End Date, Total Days Taken

3.2 Relationship design

The primary relationship described in the source is a one-to-many relationship from the employee master table to the leave transaction table using EmployeeID. One employee may have multiple leave requests, while each leave request should reference one employee record.

From table	To table	Key	Expected cardinality
Raw_Employees	Leave Requests	EmployeeID	One-to-many (1:*)

3.3 Manager hierarchy

ManagerID can be used to represent the reporting structure within the employee master table. In a typical implementation, ManagerID points to the EmployeeID of another employee who acts as the manager. This is a self-referencing hierarchy and should be validated carefully to avoid invalid manager references, loops or ambiguous paths.

3.4 Modelling checks

- Confirm that EmployeeID is unique in Raw_Employees.
- Check that every nonblank EmployeeID in Leave Requests exists in Raw_Employees.
- Confirm that RequestID uniquely identifies a leave request.
- Review blank or unmatched ManagerID values and decide how top-level managers are represented.
- Use a dedicated date table if time-based analysis is required across multiple years.

Important: *The exact relationship direction and manager-hierarchy implementation should be confirmed in the actual Power BI model. The source outline notes that single-direction or bi-directional filtering may be used depending on operational constraints.*

4. Data Cleansing & Power Query Transformations

Raw data may contain inconsistent text, duplicate records, missing identifiers, mixed data types and invalid date values. Power Query is the recommended preparation layer for profiling and correcting these issues before the data is loaded into the model.

4.1 Recommended transformation sequence

Step	Transformation	Purpose
1	Inspect source columns and data types	Understand the incoming structure and detect type mismatches.
2	Trim and clean text	Remove leading/trailing spaces and non-printable characters.
3	Standardize text values	Normalize department, job title, employment status and leave-type labels.
4	Validate identifiers	Check EmployeeID, ManagerID and RequestID for blanks or duplicates.
5	Convert date fields	Set Start Date and End Date to Date data type.
6	Validate leave duration	Check that end dates are not earlier than start dates.
7	Handle missing values	Apply documented rules for blanks and unmatched references.
8	Load prepared tables	Load clean tables into the Power BI data model.

4.2 Common Power Query operations

- **Trim:** removes extra spaces at the beginning and end of text values.
- **Clean:** removes non-printable characters that can interfere with matching.
- **Change Type:** assigns appropriate data types to IDs, dates and numeric values.
- **Remove Duplicates:** removes duplicate keys only after confirming which record should be retained.
- **Replace Values:** standardizes known variations such as inconsistent department names.
- **Conditional Column:** creates validation flags or grouped categories based on business rules.

4.3 Example quality rules

A leave request should have a valid RequestID, an EmployeeID that matches the employee master, a recognized leave type, and valid Start Date and End Date values. If Total Days Taken is supplied in the source, it should be checked against the organization's leave-counting policy rather than blindly recalculated.

The source document specifically calls out inconsistent casing, trailing whitespaces and unassigned foreign keys as examples of raw-data anomalies.

5. Missing Values, Data Quality & Validation

Missing values can create ambiguous categories such as (Blank) in charts and slicers. They can also reduce the reliability of totals and make it difficult for HR users to trace a transaction back to an employee.

5.1 Handling blanks

Field	Potential issue	Recommended handling
EmployeeID	Blank or unmatched employee key	Investigate the source record; do not assign an arbitrary employee.
ManagerID	No manager assigned	Allow blank for approved top-level roles or map to a documented category.
Department	Missing department label	Validate against HR master data; use an explicit 'Unknown' category only if approved.
Leave Type	Missing or inconsistent category	Confirm the request category with the source system or HR owner.
Start Date / End Date	Missing or invalid dates	Flag the request for review; exclude from date-based calculations until resolved.

5.2 Validation checklist

- EmployeeID is unique in the employee master table.
- RequestID is unique in the leave transaction table.
- Every leave transaction has a valid employee reference.
- Start Date and End Date are valid dates and follow the expected order.
- Total Days Taken is numeric and follows the organization's leave policy.
- Department and Leave Type values use approved labels.
- Manager references do not create circular reporting relationships.
- Totals in Power BI reconcile with the cleaned source data for a selected sample.

5.3 Audit trail and refresh controls

For repeatable reporting, retain the source file or source-system extract date, record the applied transformation steps, and document any manual corrections. After each refresh, check row counts, missing-key counts and selected KPI totals. This makes it easier to identify whether a change is caused by new business activity or a change in the incoming data.

Governance reminder: *Do not silently replace missing employee identifiers or dates with invented values. Exceptions should be visible, documented and resolved by the appropriate data owner.*

6. DAX Measures & KPI Definitions

DAX measures can provide consistent calculations for cards, charts and tables. The following examples are illustrative starting points based on the field names described in the project outline. They must be checked against the actual model, data types and leave policy before use.

6.1 Example measures

Employee Count

Counts unique employee identifiers in the employee master table.

```
Distinct Employee Count = DISTINCTCOUNT(Raw_Employees[EmployeeID])
```

Total Leave Days

Sums the supplied leave-day values in the leave transaction table.

```
Total Leave Days = SUM('Leave Requests'[Total Days Taken])
```

Leave Request Count

Counts distinct leave request identifiers.

```
Leave Request Count = DISTINCTCOUNT('Leave Requests'[RequestID])
```

Average Leave Days per Request

Calculates the average supplied leave duration per request.

```
Average Leave Days = DIVIDE([Total Leave Days], [Leave Request Count], 0)
```

6.2 Suggested KPI catalogue

KPI	Definition / interpretation
Total Employees	Unique employees in the current filter context.
Total Leave Days	Sum of Total Days Taken for the selected records.
Leave Requests	Distinct count of RequestID.
Average Leave Days	Total leave days divided by leave request count.
Leave by Type	Leave days or request count grouped by Leave Type.
Leave by Department	Leave days or request count grouped by Department.

6.3 Calculation caution

The source outline does not define leave entitlement, opening balance, carry-forward, holidays, weekends, half-days or approval-status rules. Therefore, remaining leave balance and entitlement utilization should only be calculated after the organization's policy rules and required source fields are confirmed.

7. Dashboard Visualizations & Interactivity

The project outline specifies three report pages and describes the use of slicers, button slicers and Power BI Edit Interactions. The following layout guidance expands those requirements into a practical report design.

7.1 Executive Summary page

- **KPI cards:** Total Employees, Total Leave Days and Leave Request Count.
- **Column chart:** leave days by department to compare leave volume across teams.
- **Donut or bar chart:** leave distribution by Leave Type.
- **Line chart:** leave activity over time, using a suitable date field and date table.
- **Slicers:** Department, Leave Type and reporting period.

7.2 Employee Details page

- Use a table or matrix to show EmployeeID, employee name, department, job title and manager.
- Include leave request details such as RequestID, Leave Type, Start Date, End Date and Total Days Taken.
- Provide employee and department filters for focused investigation.
- Use conditional formatting carefully to highlight incomplete or unusual records.

7.3 Managerial Hierarchy page

- Use a matrix to display manager and employee reporting levels.
- Show team leave totals by manager or department.
- Allow drill-down from manager to direct reports where the hierarchy is correctly modelled.
- Validate manager relationships before enabling hierarchy-based totals.

7.4 Slicers and Edit Interactions

Slicers should filter only the visuals that are relevant to the selected field. Power BI's Edit Interactions feature can be used to configure whether a slicer filters, highlights or does not affect a particular visual. The original project outline specifically mentions custom filtering pathways between Leave Type button slicers and managerial performance charts.

7.5 Navigation and accessibility

- Use page navigation buttons with clear labels.
- Provide descriptive chart titles and meaningful axis labels.
- Keep font sizes readable and maintain sufficient contrast.
- Use tooltips to show supporting detail without overcrowding the report page.

8. Testing, Deployment & Future Roadmap

A dashboard should be tested for both numerical accuracy and interaction behaviour before it is shared with business users. Testing should cover data refresh, relationships, calculations, slicers, hierarchy navigation and reconciliation to source records.

8.1 Testing plan

Test area	Test activity	Expected outcome
Data loading	Refresh employee and leave tables	Refresh completes and expected rows are loaded.
Key integrity	Check duplicate and unmatched identifiers	Exceptions are identified and documented.
Relationships	Filter leave records by employee	Transactions are associated with the correct employee.
DAX measures	Compare selected totals with source records	KPI results reconcile for the test sample.
Slicers	Select department and leave type	Relevant visuals respond as configured.
Hierarchy	Expand manager and direct-report levels	Reporting structure displays without circular paths.
Usability	Review titles, labels and page navigation	Users can interpret and navigate the report.

8.2 Refresh and release checklist

- Confirm the data source path and refresh credentials.
- Refresh the model and review any query errors.
- Check key KPI values and row counts after refresh.
- Test slicers and interactions on all three report pages.
- Confirm that access to employee-level information is restricted to authorized users.
- Publish the approved report and communicate its refresh frequency and ownership.

8.3 Future enhancements

- **Predictive absenteeism analysis:** explore historical patterns to identify periods of higher leave activity, subject to sufficient and appropriate data.
- **Automated threshold alerts:** notify designated HR users when agreed leave thresholds are reached.
- **Approval workflow integration:** connect request status and approval timestamps if those fields become available.
- **Policy-aware balance calculations:** incorporate entitlement, carry-forward and holiday rules after validation with HR.
- **Enhanced security:** consider row-level security where managers should see only their permitted employee groups.

9. Conclusion

The HR Leave Management Dashboard provides a structured approach to turning employee and leave transaction data into operational reporting. By combining a clear data model, repeatable Power Query transformations, consistent DAX measures and stakeholder-focused report pages, the solution can support HR review, department-level analysis and executive visibility. Ongoing data validation, access control and policy alignment are essential to keeping the report reliable and audit-ready.

Prepared as an expanded edition of the supplied five-page project outline. Illustrative calculations and recommendations should be validated against the actual Power BI implementation.